Ashwath P

Chennai, Tamil Nadu

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Education

SRM Institute of Science and Technology **Sep. 2022 – May 2026**
B.Tech – Computer Science Engineering (AI & ML) | CGPA: 9.43/10 *Chennai, Tamil Nadu*

Experience

Nokia Solutions and Networks **Oct 2025 – March 2026**
Research and Development Intern *Tamil Nadu*

- Built AI/ML computer vision pipelines for industrial automation — performing annotation, preprocessing, augmentation, and model validation to improve defect detection accuracy and manufacturing process reliability.
- Developed Industry 4.0 applications integrating React, TypeScript, Node.js, MySQL, and Raspberry Pi to enable automated process control, real-time production monitoring, and error-proof smart manufacturing operations.
- Applied supervised ML, PLC automation, and real-time validation workflows to reduce manual intervention and setup defects; collaborated with cross-functional teams on full-stack development and embedded systems integration.

Projects

Real-Time Face Anti-Spoofing System | *TensorFlow, MiniFASNet, Fourier Features, RetinaFace*

- Trained a spoof-detection model achieving 98.55% accuracy and AUC 0.98 via rigorous data preprocessing, feature engineering, and hyperparameter tuning for production-grade deployment.
- Optimized the inference pipeline to under 100ms latency through model quantization, efficient data handling, and streamlined API integration for real-time prediction throughput.

AI-Powered Resume Analyzer | *Python, Streamlit, Sentence-Transformers, Pandas, Plotly, PyPDF2*

- Built an ATS scoring engine using MPNet-v2 embeddings across 5 key dimensions, enabling intelligent resume-to-JD matching with actionable, role-specific candidate feedback.
- Implemented multi-format parsing (PDF, DOCX, TXT) with semantic NLP and interactive Plotly dashboards to deliver real-time ATS analytics and performance insights.

ADVIS – Automated Deep Visual Inspection System | *Detectron2, PyTorch, OpenCV, XGBoost, LSTM, Streamlit*

- Implemented an end-to-end vehicle damage detection pipeline processing multi-modal inputs images, sensor data, and vehicle metadata through cleaning, normalization, encoding, and feature scaling.
- Engineered a hybrid XGBoost-LSTM model with FCM and GMM clustering for accurate damage classification; delivered insights via dynamic dashboards, automated PDF reports, and real-time visualizations.

Technical Skills

Languages: Python, C, SQL
Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, Streamlit, Plotly, Sentence-Transformers, Detectron2
Libraries: NumPy, Pandas, OpenCV, Matplotlib, Seaborn, PyPDF2, pdfplumber
Developer Tools: GitHub, VS Code, Google Colab, Jupyter, REST APIs, Vercel

Research Work & Publications

Accuracy Improvement in DR using ML | *CNN, LSTM, Transfer Learning* **IEEE ICRASET 2024**

- Designed a multi-class DR severity classifier (No DR, Mild, Severe) by benchmarking CNN, Transfer Learning, CNN+LSTM Ensemble, and GBM models for robust clinical decision support.
- Achieved 97% precision, 96.8% accuracy, 96.9% recall, and 96.5% F1-score, validating model effectiveness through extensive experimentation and rigorous comparative analysis.

Certifications

Python · Neural Networks · AI Foundations · Natural Language Processing · Machine Learning · Generative AI

Leadership / Extracurricular

RBI Staff Quarters School **Jun 2024 – Jul 2024**
Data Analyst & Instructor (Volunteer)

- Performed EDA on academic datasets using statistical methods and visualization tools to extract actionable insights; taught Computer Science, English, and Mathematics with adaptive lesson plans.